
Working As Intended

A short story by Noel Gorelick,
loosely based on the article [Relativity of superluminal observers in 1 + 3 spacetime](#).

The first sign of it was a number that was too good.

I reconciled residuals back then for a reinsurance desk. Mostly relativistic long-haul freight runs, the ones that pushed close enough to c that the underwriters priced them as their own animal. You learn, doing that work, that a model is supposed to be a little bit wrong. Reality has a texture the math can't catch, and the gap between them is where the work lives. A residual of zero is not a triumph. It means you've buried the real error under a bigger one and haven't found it yet.

So when the Ceres–Vesta corridor fit to four decimal places, I filed a ticket. The logged transits sat exactly where the model had put them. And it had put them there before a single voyage came in to be counted. Clock drift, I figured. The relativistic fleets keep famously ugly timestamps, and half of actuarial science out there is arguing about whose proper second we're billing in. I flagged the corridor, sent it down to engineering, went to lunch.

It came back the next morning marked: *Working as intended*.

So I re-pulled it, and when that gave me nothing I rebuilt the timestamps off raw telemetry with the sync layer stripped out. If the clocks were lying, I wanted to catch them at it. The clocks weren't lying. The transits were genuinely short. Not reported early, not rounded into looking short. Crossed in less time than the run allowed. Which left the obvious explanation: the ships must have simply run hot, burned harder and shaved the clock. The fuel logs killed that too. They came in low to the same four decimals, the consumption of a voyage with less distance in it than we had charted. A ship that runs hot burns more. These had burned less.

It took most of three weeks to clear away everything that wasn't interesting. The rounding. A transcription fault in one of the feeds. A colleague named Pohl I was briefly sure had fat-fingered a decimal, who took the accusation better than I deserved. None of it was the thing. That's most of what the work actually is, and nobody warns you going in: weeks of clearing pigeons off the dish so you can hear the one sound that won't go quiet.

Then I pulled the rest of the corridors. It was in all of them, faint, and it had a shape. Below about half light it vanished. Near the limit it bloomed. A clerical error doesn't care how fast you're

going. But a velocity dependence is the kind of thing that makes you stop being annoyed and start being careful.

I found the paper the way you find everything, late, on a bad search. An old one from 2020, arguing that if you kept the superluminal half of the Lorentz transformation, the half everybody throws away, you got something that looked uncannily like quantum behavior falling out of plain relativity. Indeterminacy. Many paths. It had been more or less laughed off; there was a published Comment taking it apart, and the consensus, as far as I could find a consensus, was *cute, but no*. The kind of thing you cite to show you've read widely and then never use.

I emailed a physicist. She was kind about it. She explained, patiently, that superluminal frames had been ruled out, that the four-dimensional case didn't close into a symmetry group and that I should really read the Comment. I had read it and it was right. It was also completely uninterested in my corridors, which went on fitting to four decimals regardless of having been refuted. I have thought about that since. How little the data cared.

So I stopped trying to explain it and considered each anomaly a sounding off an unseen bottom, nearly useless alone, the way one cast of the lead line tells a sailor what's under him and nothing of what's around him. But I had thousands of soundings scattered among millions of records. Discovered and plotted together, they showed a floor, a surface nobody had charted because nobody had thought to look for one. It wasn't made of rock, or even of space. It was made of time, and I was plumbing it in three dimensions. Forward wasn't a line down there. It was a space, and *later* was a heading in it, the way north is a heading. You knew when you'd sounded one, because over it the order of events came loose. Not chaotic, just unfixed, like two cards that could be dealt either way and the deck not minding which came first.

I figured out that what I'd found wasn't a message. You couldn't do anything with it. I tried. The model called a run early once, weeks before the ship even cleared dock, so I priced the gain in and waited to collect. It came in on time. Ordinary time, the time the model would have given before any of this started. The foreknowledge only ever held for the runs I left alone. The instant I reached for the wheel, the road went ordinary. There is no posting a letter through a place where the only mail that gets through is the kind you didn't mean to send.

I kept at it anyway. Not for any theory, and plainly not for profit; I am the kind of person who, handed a scatter of points, can't sleep until I've seen their shape. That turned out to be the whole of the discipline, in the end: a person who couldn't leave a point unlabeled.

I left insurance, or it left me. A grant came, the desk became a footnote, and the points became the work. It took years, and I won't pretend they were exciting. Mapping is the most patient thing a person can do that isn't farming.

I learned the ground by getting it wrong over and over and writing down where. The crossings weren't beacons. Some held steady enough to chart and some flickered, and you had to decide, as craft, whether a thing that lasted a season was real enough to put ink to. I developed conventions. A notation for a crossing that closed cleanly versus one that frayed at the lip. A projection that didn't lie too badly about the temporal axes, which is harder than it sounds, because in those places there is more time than there is room, and a flat page is only ever a compromise with a geometry that has three of one thing and one of the other and no patience for paper.

The name came later. It started as a joke on a grant line. I needed a heading for the budget and "anomaly mapping" had been rejected twice. So I wrote *Temporal Cartography*, and the joke set, the way jokes do when they turn out to be load-bearing.

I will tell you the thing I am least proud of and most certain about: I never understood it. I could not have told you the mechanism if you'd held my charts hostage. I knew the texture of the terrain the way an old sailor knows a coast he could never chart; every shoal and set of the current, with no theory of tides at all. The physicists never got their mechanism either. They wrote careful papers about why it shouldn't work, but meanwhile the ships that loaded my charts came in with fewer losses, shorter transits and fewer of the small unaccountable incidents that the long runs used to shrug off as the cost of going fast. Nobody could say why. Everybody did it anyway. *It just runs better with the Atlas*. I heard that from captains who would have been insulted to be told they were navigating a disputed appendix to a refuted paper. They weren't navigating a theory. They were navigating a map, and all maps are wrong, but some are useful, the way the compass was for four centuries before anyone could explain it.

So the Atlas spread. Mine, then mine-with-additions, then editions I half-recognized. That is how a working chart lives: it gets copied, and the copies get corrected, and the correction is the highest compliment the work can receive. Every annotation in someone else's hand was proof the place was real, that another person had stood at that crossing and found my line a few meters off and fixed it.

That should have been the end of the comfortable version of the story. A woman makes a useful book she doesn't understand, and the book outlives her, which is all any of us can ask.

The annotations I didn't recognize started in the eighth edition.

I assumed a copyist. So I went looking for one: pulled the edition history, checked who had touched the plates, matched the new marks against every hand that had ever worked the Atlas. None of them. And no copyist writes your *t* the way you write it yourself, with the little drag on the crossbar I have never managed to stop and never tried to teach.

The corrections were good. That was what kept me far past the hour I should have gone home. They weren't vandalism and they weren't noise. They were *better than mine*. Then I found one marked that I had not charted yet. A crossing out past the Hilda group that I wouldn't survey for another fourteen months, drawn in already in my hand, a margin note in my conventions correcting a measurement I had not yet taken wrongly.

I understood it the only way I ever understood anything out there, by texture. *Every map is evidence that someone got there first*. You don't chart a coast that no one has reached; the chart *is* the reaching, written down. I had been treating "first" as a fact, but it's only the direction your time happens to point, and out past the Hildas there is more than one. Across a fold the order of who-wrote-what comes loose along with everything else.

Nothing was reaching me from the future. The Atlas was one consistent document, consistent because the geometry will not carry any other kind, and it was being written, all of it, by a cartographer spread along more directions of time than a page can hold, who can't agree with herself on the order she did it in and doesn't have to. The annotations in my hand were mine. The ones I had made. The ones I would make. The ones, past some crossing I'll come to when I am old, that I had already made and forgotten in advance.

I couldn't ask it anything. I tried. I left a question in the margin, a thing only I would know to ask. Nothing came back. The wheel goes ordinary the instant you reach for it, and that law held all the way.

The question was gone by morning. Where it had been there was a mark in my hand, and it wasn't an answer, it was a correction. Years back I'd named a crossing for my mother and spelled her the way I always have, the way every form and every keyboard spells her — Zoe. The accent never survived anything; I'd stopped seeing it gone by the time I was ten. Whoever was further along had put it back. Zoë.

I picked up the pen. I have corrections of my own to make, for a woman further along who writes her *t* the way I do and is going to need them. The work continues. It always did, in every direction. That is the only part I am sure of.

Working as intended.